

Q1.

I think that it would be unlikely that the flattened 1D testing image would be correctly identified by the feedforward neural network, since the FNN cannot identify the spatial structure of the image. The 2D input using convolution and pooling layers will be much more likely to identify the image since it can actually use the spatial structure

Q2.

Layer 1: Each Kernel is $5 \times 5 = 25$. There are 6 kernels for each input, 1 input, and 12 bias terms.

$$25 \times 6 \times 1 + 12 = 162$$

Layer 2: Each Kernel is $12 \times 12 = 144$. There are 12 kernels for each input, 6 inputs, and 12 bias terms.

$$144 \times 12 \times 6 + 12 = 10380$$

$$162 + 10380 = \mathbf{10542}$$

Q3.

A. $(\text{image} - \text{mean}) / \text{std}$

$$\text{Max: } (1 - 0.5) / 0.5 = \mathbf{1}$$

$$\text{Min: } (0 - 0.5) / 0.5 = \mathbf{-1}$$

B. $\text{Max: } 1 - 0/1 = \mathbf{1}$

$$\text{Min } 0 - 0/1 = \mathbf{0}$$

Q4.

A.

Red:

$$\text{Max: } (1 - 0.49139968) / 0.24703233 = 2.059$$

$$\text{Min: } (0 - 0.49139968) / 0.24703233 = -1.989$$

Green:

$$\text{Max: } (1 - 0.48215827) / 0.24348505 = 2.127$$

$$\text{Min: } (0 - 0.48215827) / 0.24348505 = -1.98$$

Blue:

$$\text{Max: } (1 - 0.44653124) / 0.26158768 = 2.116$$

$$\text{Min: } (0 - 0.44653124) / 0.26158768 = -1.707$$

B.

I would not expect this model to perform well on the test data in this scenario since the test data uses inputs in range $[0, 1]$, while the test data had a range of $[\sim 0.24, \sim 0.49]$.

Because the train and test data are normalized differently, I think the model would not be accurate.

Q5.

Crowbar, lockpicking tools, screwdrivers, ability to disable alarms, ability to stay quiet

Q6.

- A. This would be considered a successful attack if the attacker were to attach a seemingly normal link to the website that students would click and enter personal information into, such as bank account information. The attacker could make money from this.
- B. The attacker could use the users personal files as ransomware, only agreeing to decrypt the files if the user pays them money.
- C. If the website has weak enough authentication, an attacker could use SQL injection to access the database with the drivers license and use them to commit financial fraud by opening credit cards under the hacked user's identities.

Q7.

```
(pytorch-env) maya_geva_uri_edu@gypsum-gpu004:~$ python /home/maya_geva_uri_edu/C5C592/Assignment4/F05MLab/F05MLab.py
CIFAR-10 Clean Val Loader Acc: 0.9278
CIFAR-10 Clean Correct Loader Acc: 1.0
Processing up to sample= 64
Processing up to sample= 100
CIFAR-10 F05M Loader Acc: 0.25
(pytorch-env) maya_geva_uri_edu@gypsum-gpu004:~$
```

Q8.

```
# Calculate the loss

loss = torch.nn.CrossEntropyLoss()
```

The loss function is not soft coded. It is calculated using the cross entropy loss function.

Q9.

```
perturbedImage = xData + epsilonMax*xData.grad.data.sign().detach()
```

Q10.

```
epsilonMax = 0.031 #Maximum perturbation
```